

ISLAMIC UNIVERSITY OF TECHNOLOGY



CSE 4531

E-COMMERCE AND WEB SECURITY

SEO & Web Analytics

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11 March 2026

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1 Introduction

This report documents our hands-on implementation of Search Engine Optimization (SEO). We built a live website and recorded real traffic data before making any changes. Afterwards, we applied some well-known SEO techniques, and then measured the outcome directly through *Google Analytics* 4.

The site was built around a deliberately light-hearted topic: **Pigeons vs Artificial Intelligence**, chosen because it enabled us to target specific, low-competition keywords, hence making it realistic to observe measurable results within a short time span. The site was deployed publicly at <https://nashatislam.github.io/ecws-dummy/> and remained live throughout the observation period.

2 Objectives

The project had three core objectives:

- **Search Visibility:** To ensure the site could be discovered and indexed by search engines through correct on-page SEO implementation.
- **Credibility & Trust:** To improve perceived authority by applying structured headings, canonical URLs, descriptive meta tags, and proper sitemaps.
- **Sustainable Traffic:** To generate organic, long-term traffic without having to rely on paid advertising.

3 Methodology

The project followed a workflow containing four distinct phases:

3.1 Phase 1: Site Building and Deployment

A two-page HTML/CSS website was built without using any framework. The site comprised:

- `index.html`: The homepage presenting the '**Pigeons vs AI**' concept.

- `blog.html`: a detailed article describing a few things pigeons can do that AI still cannot.

It was then deployed publicly via *GitHub Pages*.

3.2 Phase 2: *GA4* Baseline Recording

Before any SEO changes were applied, *Google Analytics 4 (GA4)* was embedded on both pages using a standard `gtag.js` script supplied by the Tag Manager in *GA4*. Baseline traffic data was recorded over 3 days. During this period, the site collected the following:

Metrics	Baseline Values
Views	21
Active Users	7
Total Users	8
Sessions	10
Average Session Duration	1m 48s
Bounce Rate	30.0%
Traffic Sources	Direct only

Table 1: GA4 baseline metrics recorded before SEO implementation

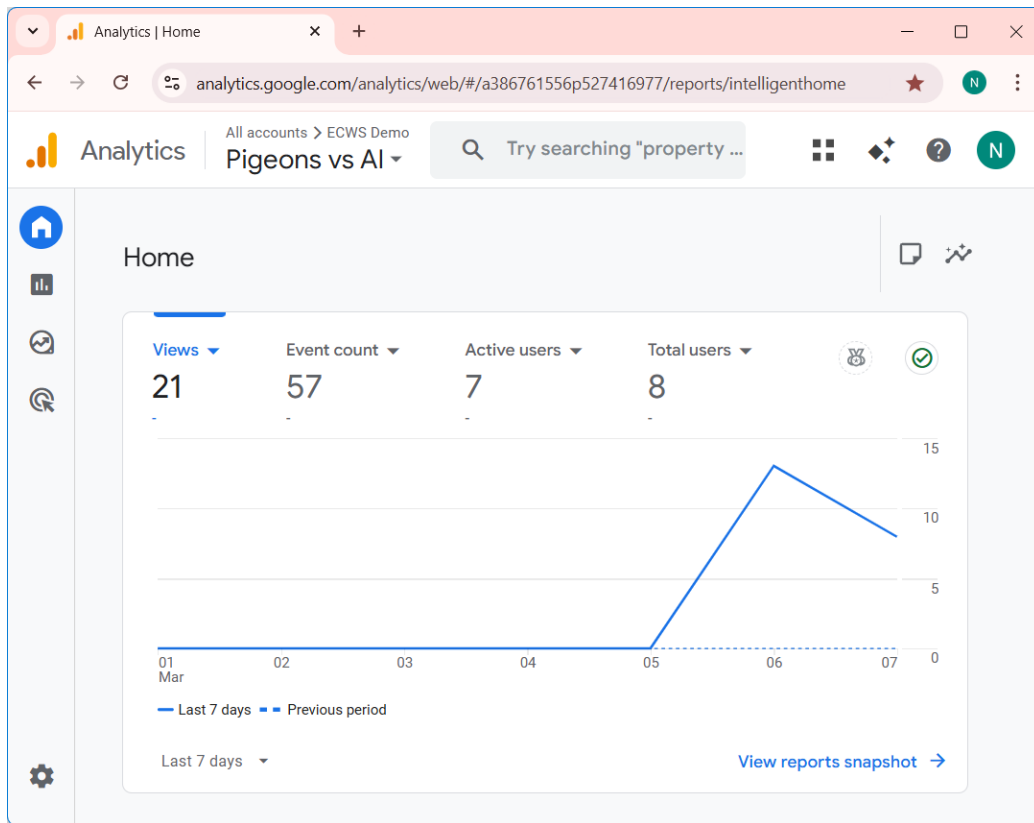


Figure 1: Views pre-SEO

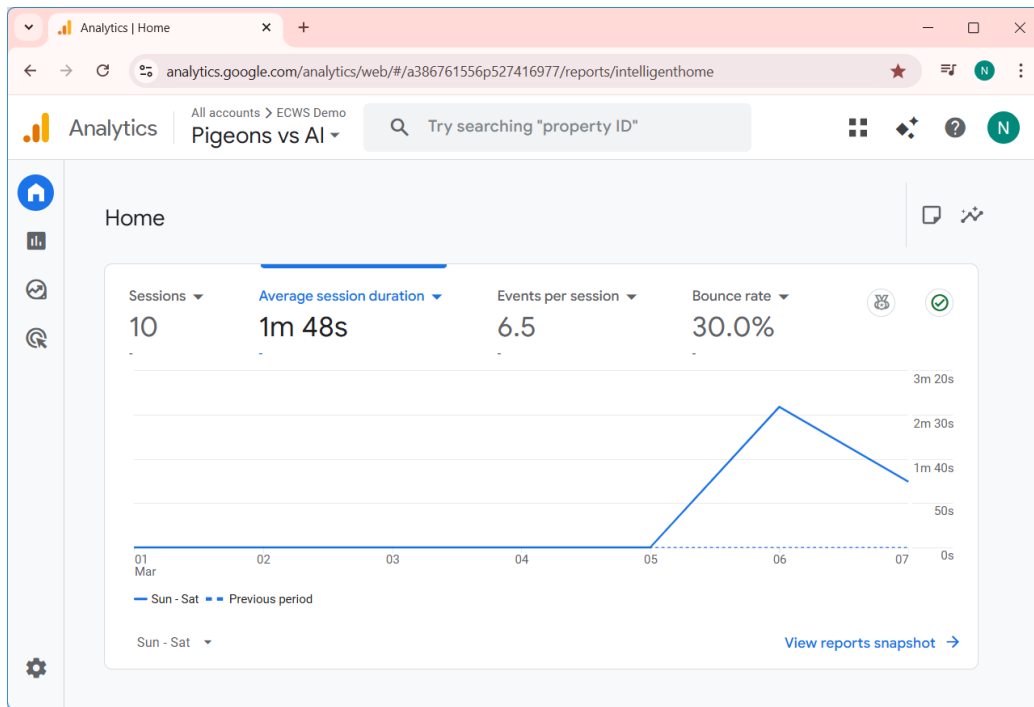


Figure 2: Session duration pre-SEO

3.3 Phase 3: SEO Implementation

On March 8, 2026, a comprehensive set of SEO elements was applied across both pages.

3.3.1 Title Tags

Descriptive title tags were added to both pages, containing relevant keywords:

```

1 <!-- index.html -->
2 <title>Pigeons vs AI: Who Is Really Smarter?</title>
3
4 <!-- blog.html -->
5 <title>5 Things Pigeons Can Do That AI Still Can't | Pigeons
   vs AI</title>

```

3.3.2 Meta Descriptions

Concise, keyword-integrated meta descriptions were written for each page, both falling within Google’s recommended 155–160 character limit.

```
1 <!-- index.html -->
2 <meta name="description" content="A fun deep-dive into pigeon
3 intelligence, AI limitations, and why AI vs animals is a more
4 interesting debate than you think." />
```

3.3.3 Heading Hierarchy (H1/H2)

A strict heading structure was implemented across both pages:

- Each page uses a single H1 tag, which contains the primary keyword.
- Multiple H2 tags split up the content into clearly defined, relevant sections.
- This hierarchy indicates content structure to search engine crawlers and improves readability scores.

3.3.4 Keyword Placement

Target keywords were embedded in various locations across both pages:

Element	Detail
Primary keyword	Pigeons vs AI
Secondary keywords	AI vs Animals, Pigeon Intelligence, AI Limitations, Pigeon vs Artificial Intelligence
Placement locations	Title tags, meta descriptions, H1/H2 headings, introductory paragraphs, conclusion sections

Table 2: Keyword placement strategy

3.3.5 Canonical URLs

Canonical link elements were added to prevent content duplication issues:

```

1 <!-- index.html -->
2 <link rel="canonical"
3   href="https://nashatislam.github.io/ecws-dummy/index.html"
4   />
5 <!-- blog.html -->
6 <link rel="canonical"
7   href="https://nashatislam.github.io/ecws-dummy/blog.html" /
8   >

```

3.3.6 Robots Meta Tag

Both pages include the robots directives, to explicitly instruct search engine crawlers to index the pages and follow all outbound links:

```

1 <meta name="robots" content="index, follow" />

```

3.3.7 XML Sitemap

A sitemap.xml was created and linked within both HTML pages:

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9">
3   <url>
4     <loc>https://nashatislam.github.io/ecws-dummy/index.html<
5     /loc>
6     <lastmod>2026-03-05</lastmod>
7     <changefreq>monthly</changefreq>
8     <priority>1.0</priority>
9   </url>
10  <url>
11    <loc>https://nashatislam.github.io/ecws-dummy/blog.html</
12    loc>
13    <lastmod>2026-03-05</lastmod>
14    <changefreq>monthly</changefreq>
15    <priority>0.8</priority>
16  </url>
17 </urlset>

```

3.3.8 ALT Text for Images

The pigeon illustration on the homepage was given a descriptive label:

```

1 <svg aria-label="Cute grey pigeon illustration" ...>

```

3.4 Phase 4: Evaluation

Following SEO implementation, the site was evaluated using three tools:

- ***Google Lighthouse***: to audit technical SEO and performance scores.
- ***HigherVisibility SERP Snippet Optimizer***: to validate meta tag rendering by the web browsers.
- ***Google Analytics 4***: to compare traffic metrics before and after implementation.

4 Key Findings & Results

4.1 *Google Lighthouse* Scores

After applying all SEO elements, the site was audited using *Google Lighthouse*. The results confirmed that every required technical SEO criterion was implemented correctly:

Metric	Score
SEO Score	100 / 100
Performance Score	97 / 100
First Contentful Paint (FCP)	0.6 s
Largest Contentful Paint (LCP)	1.0 s
Speed Index	0.7 s
Total Blocking Time (TBT)	0 ms
Cumulative Layout Shift (CLS)	0.008

Table 3: *Google Lighthouse* audit results after SEO implementation

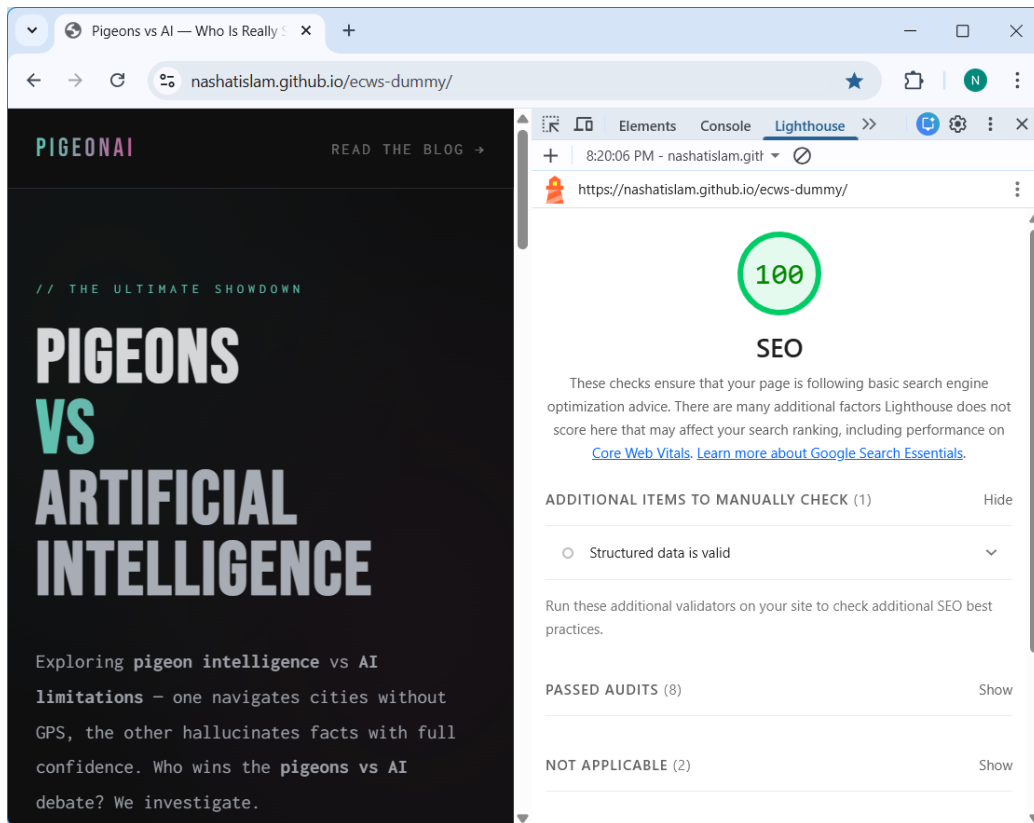


Figure 3: Perfect SEO score

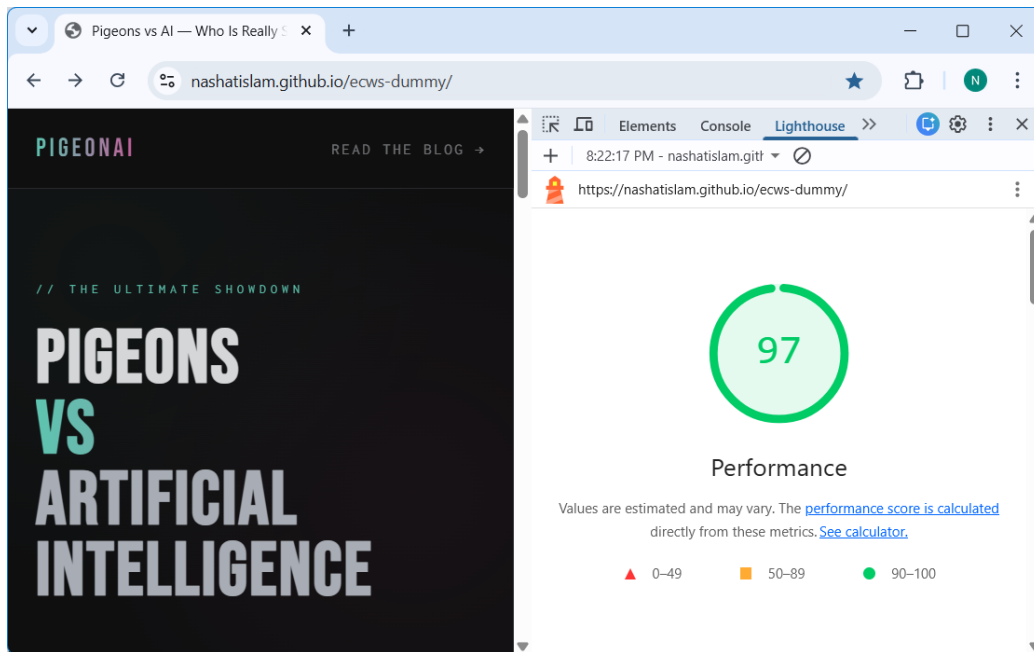


Figure 4: Near perfect performance score

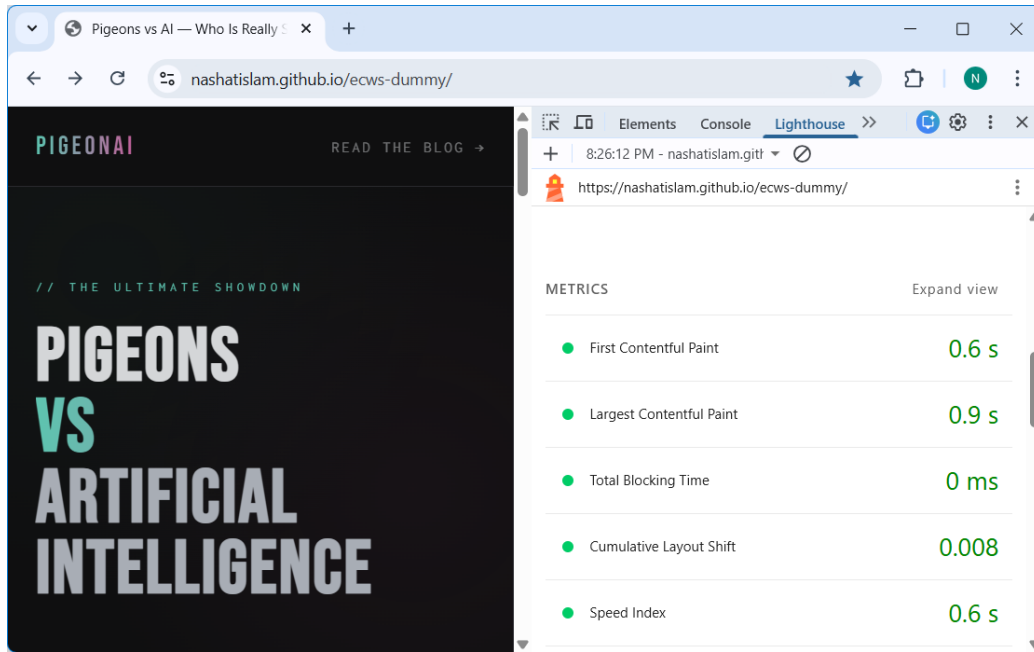


Figure 5: Performance metrics scores

Key Points:

- All 8 Lighthouse SEO audits passed, including 'page not blocked' from indexing, a valid `<title>` element, meta description present, successful HTTP status code, descriptive link text, crawlable links, valid hreflang, and valid `rel=canonical`.
- The excellent performance score confirms that SEO implementation did not compromise page load speed.

4.2 SERP Preview Validation

Using the *HigherVisibility* SERP Snippet Optimizer, the title, URL, and meta description for both pages were confirmed to render correctly within Google's recommended character limits.

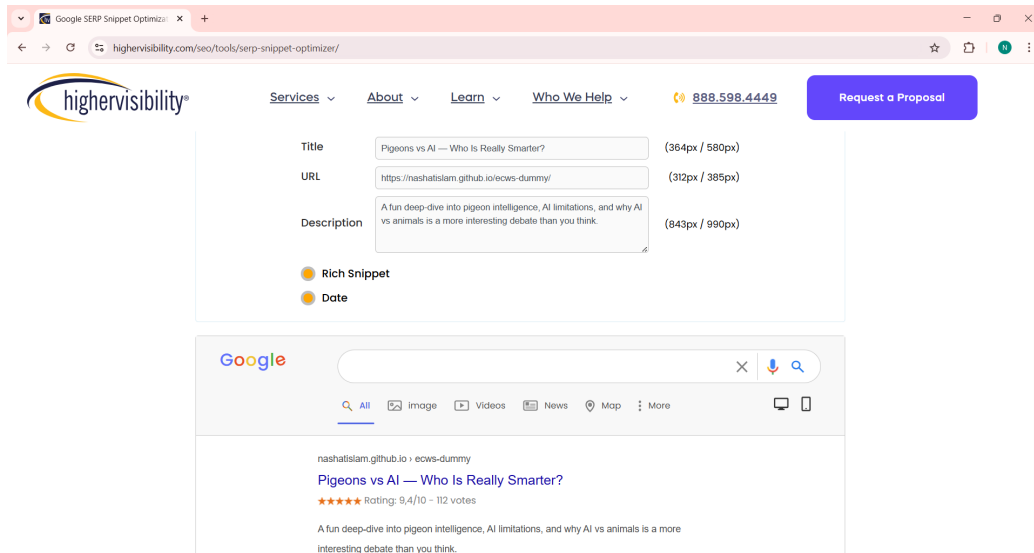


Figure 6: SERP preview

4.3 *GA4* Traffic Results

The most important findings came from *GA4*, where a clear and measurable improvement in all key metrics was observed following SEO implementation on March 8:

Metric	Before SEO	After SEO
Views	21	105
Active Users	7	20
Total Users	8	20
Sessions	10	37
Avg. Session Duration	1m 48s	5m 13s
Events per Session	6.5	7.1
Bounce Rate	30.0%	54.1%
Traffic Sources	Direct only	Direct + Organic Social

Table 4: *GA4* metrics before and after SEO implementation

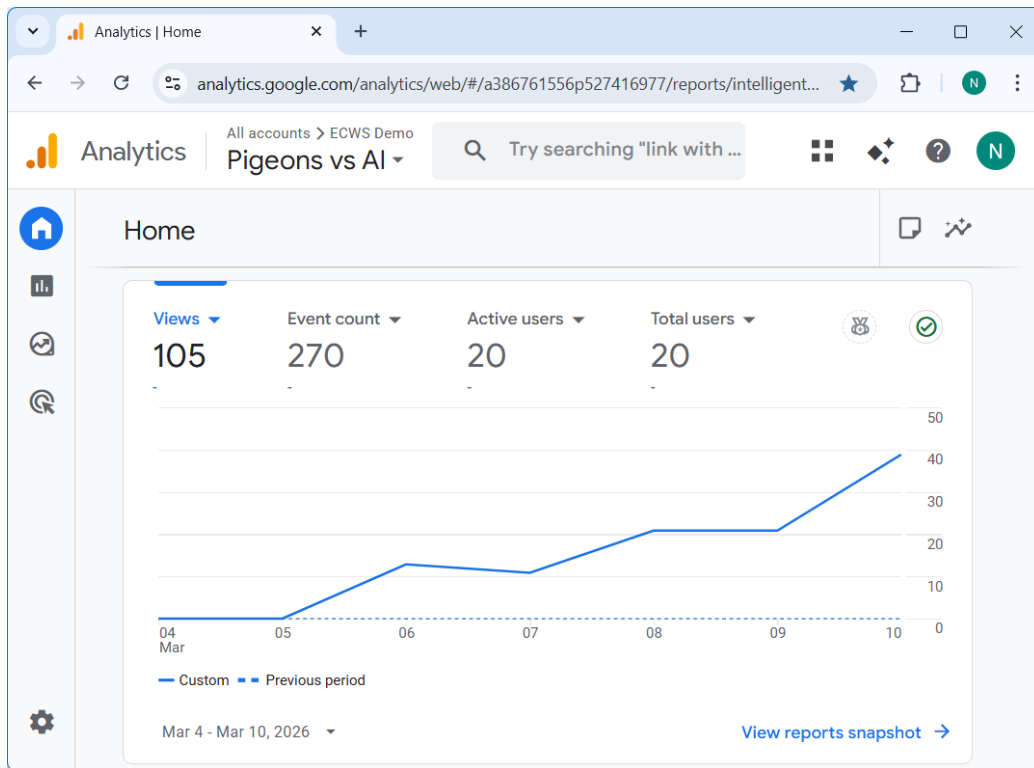


Figure 7: Views after SEO implementation

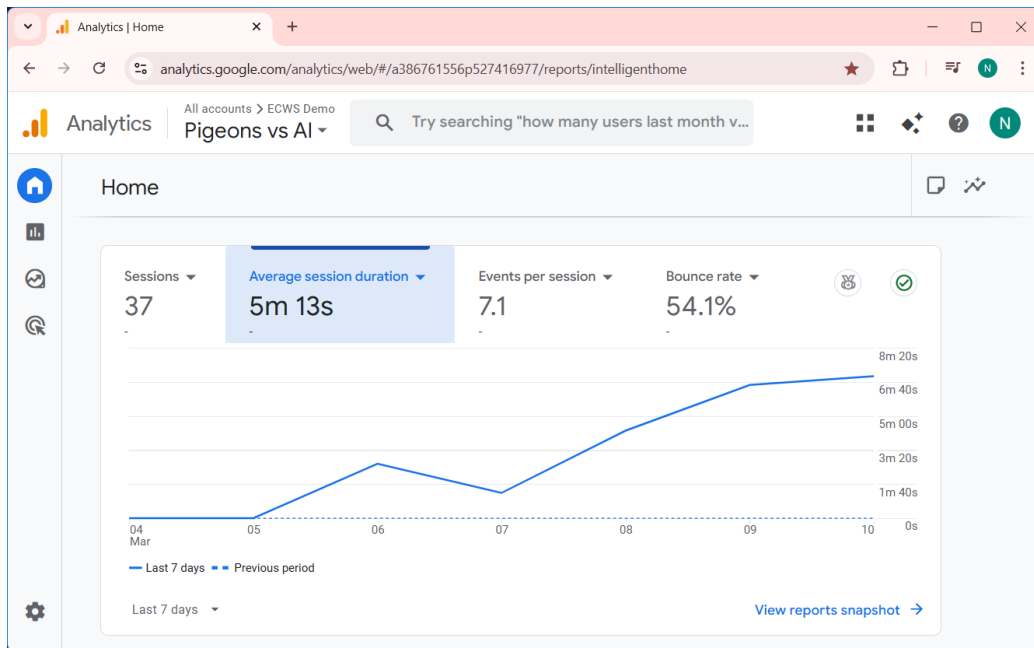


Figure 8: Session duration after SEO implementation

Key Findings:

- **Traffic Growth:** Views increased by over 380% and sessions nearly quadrupled following implementation.
- **Deeper Engagement:** Average session duration jumped from under 2 minutes to over 5 minutes, which indicates users were engaging meaningfully with the content.
- **Traffic Diversification:** Sources expanded from direct-only to include organic social, hence confirming improved discoverability.
- **Note on Bounce Rate:** The increase from 30% to 54.1% reflects the influx of new visitors from organic social sources. This is a common and expected pattern when traffic diversifies beyond a website's direct network.

5 Conclusion

The findings of this project validate the practical effectiveness of Search Engine Optimization as a valuable web development discipline. Across all observed metrics, a clear and consistent improvement was recorded following implementation. This confirms that structured SEO techniques produce tangible, data-backed results even within a short observation window.

Reflecting on the process, the most significant insight was how quickly and directly SEO choices translated into measurable outcomes. This has broader implications in an e-commerce context, where organic search visibility is a critical factor of sustainable traffic and conversions without the ongoing cost of paid advertisement.

6 References & Tools Used

- *Google Analytics 4 (GA4)* <https://analytics.google.com>
- *Google Lighthouse*
- *HigherVisibility* SERP SnippetOptimizer <https://www.highervisibility.com/seo/tools/serp-snippet-optimizer/>
- *GitHub Pages* — <https://pages.github.com>